



NEOXAM DATAHUB – DATA QUALITY CONTROL NEW DESIGN

DATA MANAGEMENT ADMINISTRATION



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Introduction

The new Data Quality Control module enables setting-up controls that mix different fields and types of data control within an execution context of Data Quality Control.

The new model is based on two main elements:

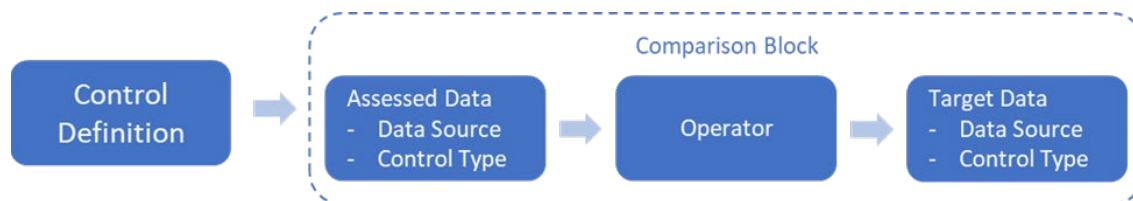
- **Data control:** defines a set of data verification over different data values
- **Control Execution rules:** define how the controls will be triggered and for what scope of objects.

Control Definition

Overview

A control defines a set of verification to be applied over data values. Each control is built for a specific type of data structure: Business Dictionary, Data Dictionary or Scope.

Several verifications can be run within a control; therefore, the main component of a control is the comparison block, which represents the verifications to be executed within a control. Within this block, comparison clauses are defined by comparing Assessed Data against Target Data.



Business Class Control Creation and Set-Up

Currently, the business classes controls can be set-up on the screen **Datahub Factory > Data Administration > Data Quality > DQC > Data Controls**. Each control is created having as checking scope a specific business class.

To configure a new control:

1. Open the **Data Controls** screen.
2. Click the **Add Control** action button.

It enables creating a new control for the class selected in the grid.



Note: If the action **Add Control** is not available because any control is selected, a new control can be added directly from the grid action “+”. In that case, an empty control is added and all information must be filled out manually.

3. From the **Control creation** window, enter the following information:

- Control name
- Quality indication
- Quality level

Control creation:

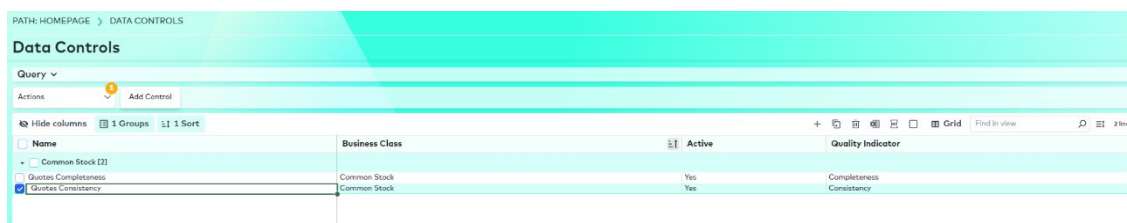
Control name:
Quotes Consistency

Quality Indicator:
Consistency

Quality Level:
Suspect data

Ok Cancel

4. Click **Ok** to save.
The new control is created.



Control Customization

Once the control is created, you need to customize it by defining additional set-up parameters with the **Control Detail screen**.

PATH: [HOMEPAGE](#) > [DATA CONTROLS](#) > [CONTROL DETAILS: EQUITY FIELDS COMPLETENESS](#)

Control Details: Equity Fields Completeness

Actions

Identification

Name(en)	Active
Equity Fields Completeness	☒
Quality Level	Quality Indicator
	Completeness
Comment	

Applicability/Groups

Applicability	Business Class
Business Dictionary	Equity

Data Check Clauses

7 Hidden columns
 Group
 1 Sort
 +
 -
 Grid
 Find in view
 4 lines

☐	Clause
☒	Value of ISIN Is Null
☐	Value of BBG Unique ID Is Null
☐	Value of CUSIP (USA) Is Null
☐	Value of BBG - Security Type 1 Is Null

Identification

- **Name:** Control name
- **Active:** Flag to indicate if control is active. Only active controls are checked during quality screening.
- **Quality Level:** qualifies the data at the origin on a data issue and is used for golden copy calculation rules.
- **Quality Indicator:** indicates the data quality indicator impacted by a data issue found.
- **Comment:** Generic comment to help user understand the scope of verification being executed.



Applicability

- **Applicability:** determines if the control is to be applied on Data Dictionary, Business Dictionary or Scopes (only Business Dictionary is currently available).
- **Business Class:** business class in scope of current control. Only fields from selected business class will be available for clauses construction.

Adding Control Clauses

Comparison clauses can be added and deleted using the **Add**, **Duplicate**, **Delete** buttons available grid actions.

PATH: [HOME](#)PAGE > DATA CONTROLS > CONTROL DETAILS: EQUITY FIELDS COMPLETENESS

Control Details: Equity Fields Completeness

Q Actions

Identification

Name(en)
Equity Fields Completeness

Active
☒

Quality Level
▼

Quality Indicator
Completeness ▼

Comment

Applicability/Groups

Applicability *
Business Dictionary ▼

Business Class
Equity ▼

Data Check Clauses

7 Hidden columns Group 1 Sort

+ □ ✕ 📄 🗑️ 🔍 Find in view 🔍 4 lines

Clause

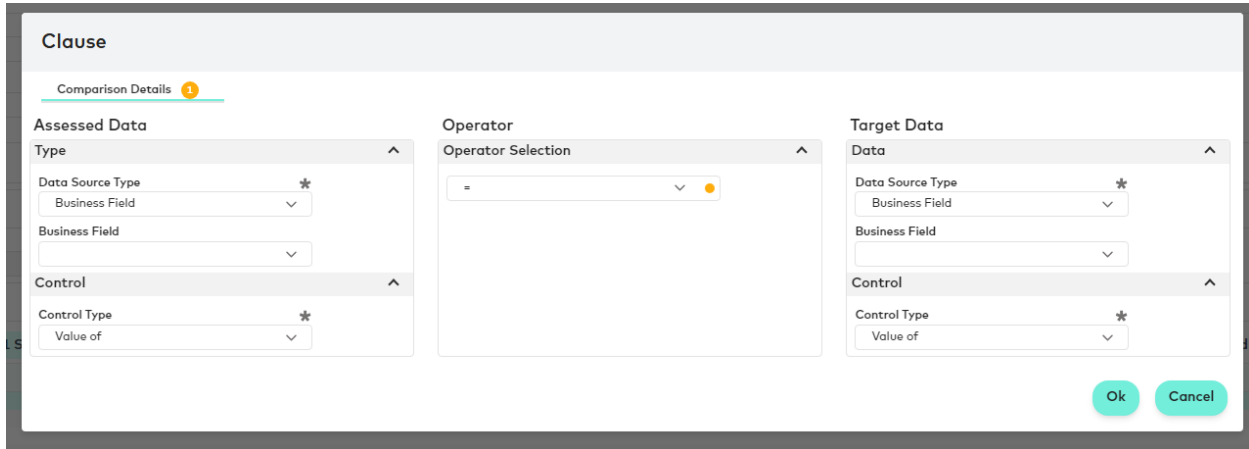
☒ Value of ISIN Is Null

☐ Value of BBG Unique ID Is Null

☐ Value of CUSIP (USA) Is Null

☐ Value of BBG - Security Type 1 Is Null

The **Add** icon (+) displays the following pop-up:



You need to define the **Assessed Data**, **Operator** and **Target Data** blocks.

Both **Assessed Data** and **Target Data** are defined as follows:

- **Data source:** defines the data source being verified. Can be a Business Field Value of a Fixed Value.
- **Control Type:** control type that is applied to the Data Source. Value Of, Series Jump, Stale
 - **Value Of** returns the current value of the data source.
 - **Series Jump** returns the variation between the current value of the data source and the previous one.
 - **Stale** returns the number of days since a value has not been updated in a Time Series.
- **Target Data** is not required in case of standalone operators (is Null, is Not Null).
- **Operator:** the list of available operators is restricted based on the Assessed Data configuration.

Below is the list of operators available depending on the Assessed Data resulting data type:

Resulting Data Type	Control Clause Available Operators
Numeric - Date	All
List - Object - String	<> ; = ; is Null ; is not Null

- **Target Data - Business Field:**

The list of available business fields is filtered to include only options whose data type matches the resulting data type of **Assessed Data**.

Comparison Block

Comparison

Assessed Data

Type

Data Source Type *

Business Field

Business Field

Activity Country

Control

Control Type *

Value of

Pin the ribbon

Operator

<>

Target Data

Type

Data Source Type *

Business Field

Business Field

Activity Country *

Country Issue

Country of Incorporation

Market Activity Country

- Target Data – Fixed Value:** the input control is adapted to the resulting data type of the Assessed Data:
 - Object:** the list of instances for the selected object is displayed

Comparison

Assessed Data

Type

Data Source Type *

Business Field

Business Field

Activity Country

Control

Control Type *

Value of

Pin the ribbon

Operator

<>

Target Data

Type

Data Source Type *

Fixed Value

Value

Afghanistan

Africa

Africa - Central

Africa - Eastern

Africa - Northern

Africa - Other Sub-Saharan

Africa - Southern

Africa - Sub-Saharan

- List:** all list items of the target list are displayed

Comparison Block

Comparison

Assessed Data

Type

Data Source Type *

Business Field

Business Field

ICB Sector

Control

Control Type *

Value of

Pin the ribbon

Operator

<>

Target Data

Type

Data Source Type *

Fixed Value

Value

OIL & GAS

Technology

Basic Materials

Telecommunications

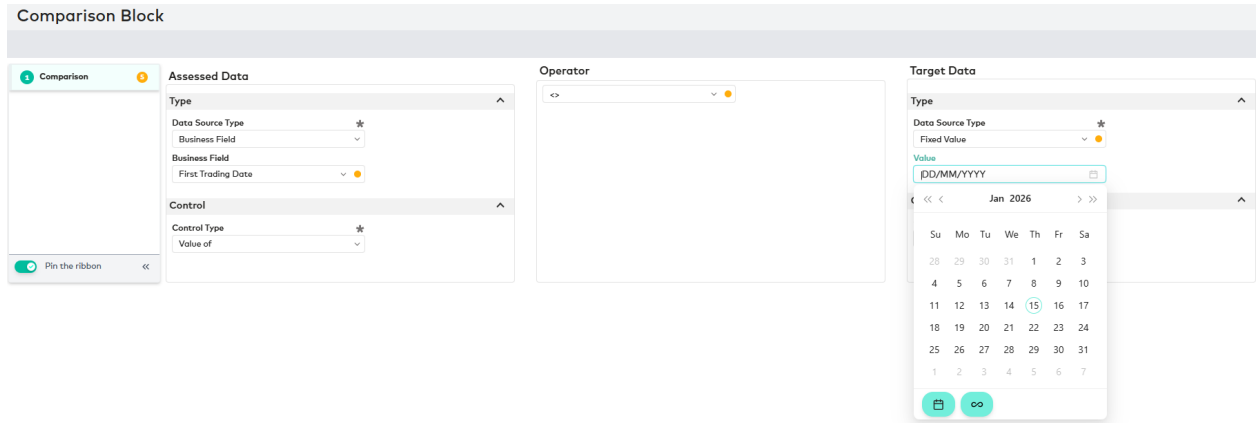
Health Care

INDUSTRIALS

Financials

Consumer Goods

- **Date:** a date picker is displayed to select a specific date.



- **Numeric:** only numeric values can be entered in the field.

Examples of Control Clause with Series Jump Control Type

The **Series Jump** control type is designed to evaluate the difference between values within a Time Series.

It applies to a business field and returns the variation between the current value and the previous one. This control type has several parameters that need to be set to define which calculation mode will be used to perform the evaluation:

- **Calculation mode:** % Difference or Value Difference

If this field is left empty, the % Difference calculation mode applies by default.

- **Factor:** decimal between 0 and 1. This field refers to the control factor to evaluate the difference between values.
If left empty, the value is 1, so that there is no impact on the computation.
- **Absolute Value:** Yes or No. This field indicates if the system returns the modulus of the calculated value. If this field is left empty, No is applied by default.
 - **When Absolute Value = No**, then a positive result indicates that the difference between values represents an increase, whereas a negative result indicates a decrease.
 - **When Absolute Value = Yes**, then a positive result will always be returned.
- **Time-depth:** This field refers to the number of days to be considered for values comparison. If left empty, 1 is considered as the default value.

Series Jump Value Difference Calculation

For **% Difference calculation** mode, the Series Jump evaluation returns the results of the following value: $(\text{current_value} - \text{previous_available_value}) * \text{Factor}$

The screenshots below show the comparison clause configuration for the control **"Series Jump - Close Quote > 1"**

Comparison Block

Comparison	Assessed Data	Operator	Target Data
Type Data Source Type * Business Field ▼ Close Quote ▼ Control Control Type * Series Jump ▼ Calculation Mode ▼ Value Difference ▼ Factor ▼ 0.90 ▼ Absolute Value ▼ Time Depth ▼	Operator > ▼	Target Data Type Data Source Type * Fixed Value ▼ Value ▼ 1.0000 ▼ Control Control Type * Value of ▼	

Pin the ribbon

Control Details: Series Jump - Close Quote > 1

Q Actions

Identification

Name(en) Active

Series Jump - Close Quote > 1 ✓

Quality Level Quality Indicator

Suspect data Accuracy

Comment

Series Jump (Close Quote) > 1

Applicability/Groups

Applicability * Business Class

Business Dictionary Equity

Data Check Clauses

+ Grid Find in view 1 lines

#	Clause
0	Series Jump Close Quote > Value of 1

Series Jump % Difference Calculation

For % Difference calculation mode, the Series Jump evaluation returns the results of the following value: $((\text{current_value} - \text{previous_available_value}) / \text{previous_available_value}) * \text{Factor}$

The screenshots below show the comparison clause configuration for the control **"Series Jump - High Quote > 10%"**

Comparison Block

Comparison	Assessed Data	Operator	Target Data
Comparison Type Data Source Type Business Field High Quote	Type Data Source Type Business Field High Quote	Type Data Source Type Fixed Value Value 0.1000	Type Data Source Type Fixed Value Value 0.1000

Pin the ribbon

Control Details: Series Jump - High Quote > 10%

Identification

Name(en) Series Jump - High Quote > 10%	Active ☒
Quality Level Suspect data	Quality Indicator Accuracy
Comment 	

Applicability/Groups

Applicability * Business Dictionary	Business Class Equity
--	--

Data Check Clauses

☒ 5 Hidden columns
 ☒ Group
 ☒ 1 Sort

#	Clause
0	Series Jump High Quote > Value of 0.1

Series Jump with Time-depth

Time-Depth parameters can be used with both calculation modes.

If a value (e.g D-3) is selected, then the previous value is calculated as follows:

$$\text{previous_available_value} = \text{average}(\text{last_available_value over the number of days in Time_depth}).$$

The screenshots below show the comparison clause configuration for the control **Series Jump - Open Quote > 0.1 D-3**

Comparison Block

Comparison	Assessed Data	Operator	Target Data
Comparison	Type Data Source Type * Business Field Open Quote Control Control Type * Series Jump Calculation Mode % Difference Factor 0.90 Absolute Value Time Depth D-3		Type Data Source Type * Fixed Value Value 0.1000 Control Control Type * Value of

Pin the ribbon

Control Details: Series Jump - Open Quote > 0.1 D-3

Q Actions

Identification

Name(en) Series Jump - Open Quote > 0.1 D-3 Active ☒

Quality Level Suspect data Quality Indicator Accuracy

Comment Series Jump - Open Quote > 0.1

Applicability/Groups

Applicability Business Dictionary Business Class Equity

Data Check Clauses

5 Hidden columns Group 1 Sort

#	Clause
0	Series Jump Open Quote > Value of 0.1

Examples of Control Clause with Stale Control Type

The **Stale** control type is designed to count the number of days since a value has not been updated within a Time Series.

This control type has no parameter.

The screenshots below show the comparison clause for the control **Check Quote Open Staleness**.

Comparison Block

Comparison	Assessed Data	Operator	Target Data
	Type Data Source Type * Business Field ▼ ● Business Field ▼ ● Open Quote ▼ ● Control Control Type * Stale ▼ ●	>	Data Data Source Type * Fixed Value ▼ ● Format Numeric ▼ ● Value 1 ▼ ● Control Control Type * Value of ▼

Control Details: Check Quote Open Staleness

Q Actions

Identification

Name(en)

Active ☒

Quality Level Suspect data ▼

Quality Indicator ▼

Comment

Applicability/Groups

Applicability * Business Dictionary ▼

Business Class Equity ▼

Data Check Clauses

🗑️ 📄 🔍 + 🗪 📋 🔍 Grid Find in view 🔍 ☰ Selected Rows 0

#	Clause
0	Stale Open Quote > Value of 1

Duplicating Control Clause

The **Duplicate** button creates a new editable comparison based on the selected comparison.



Deleting Control Clause

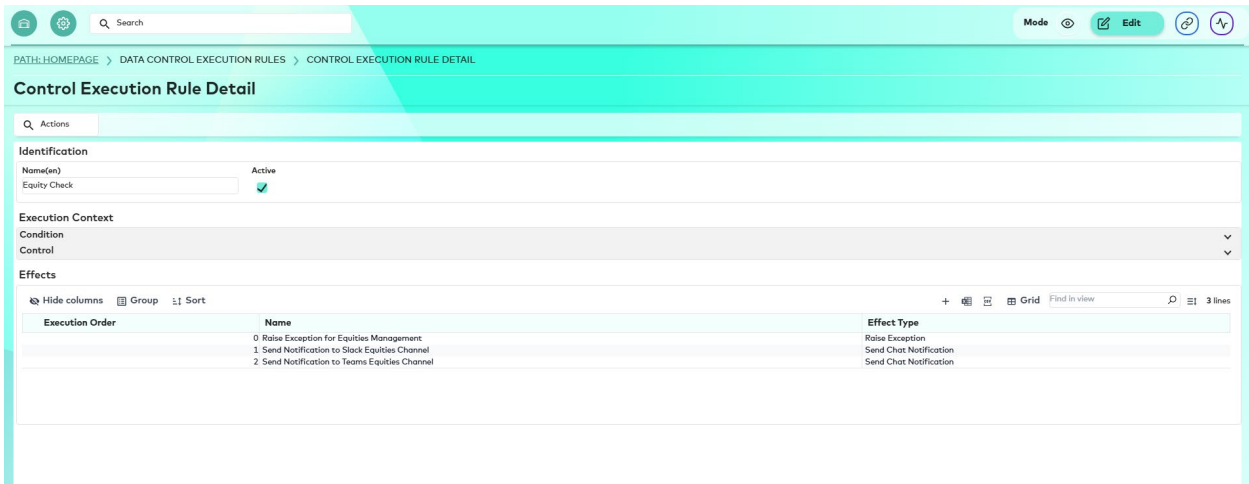
The **Delete** button can be used to delete a comparison clause. All links to the deleted clause are removed from the system (exceptions, tracking).

Rule Creation and Set-Up

The execution rules can be managed on the **Datahub Factory > Data Administration > Data Quality > DQC > Data Control Execution Rules** screen.

The goal of defining a rule is to specify the controls that must be executed given a specific applicability condition (trigger + filtering) and what effects must be applied when a data issue is found.

The following blocks are available in the detail of a rule:



Execution Order	Name	Effect Type
0	Raise Exception for Equities Management	Raise Exception
1	Send Notification to Slack Equities Channel	Send Chat Notification
2	Send Notification to Teams Equities Channel	Send Chat Notification

Identification

- **Name:** Rule name
- **Active:** Flag to indicate if a rule is active. Only active rules are executed.

Execution context

- **Condition:** defines the applicability condition of a rule, including the applicability scope (currently only business class supported) and an applicability condition. In the example below, the rule will be executed only on objects having **Activity Country** = France.

PATH: HOMEPAGE > DATA CONTROL EXECUTION RULES > CONTROL EXECUTION RULE DETAIL

Control Execution Rule Detail

Search Actions

Identification

Name(en) Active ☒

Execution Context

Condition

Applicability Business Class

Select the main object class and related classes

Hide columns Group Sort

Class name

You Can Define Object Criteria

	Field	Rec spec	Oper	Value	Category
And	Country		=	France	

+ Add criterion + Add group

View Criteria Expression

- **Control:** defines the list of controls to be checked when the rule is executed. Only active controls having the same applicability scope as the rule can be added.

Control

2 Hidden columns Group Sort

Control
Equity Prices Completeness
Equity Prices Consistency
Equity Fields Completeness

Effects

Effects

Several effects can be added as output of an execution rule when a data issue is found. Currently only the **Raise Exception** type is supported.

Effects

1 Hidden column Group Sort

Execution Order	Effect Type
	Raise Exception

Note: Business Dictionary inheritance is considered for rules execution. It means that rules applicable to a parent class will be applied for child classes. Example: rules set-up for Security Core class will be executed when quality checks are launched for an Equity.

Effects Set-Up

Raise Exception

Specific parameters can be set-up according to the effect type. For **Raise Exception**, the following information can be defined:

Control Effect Details

Q Actions

Effect

Identification

Name(en)
Raise Exception for Equities Management

Name(fr)
Créer Alerts

Code
EQ_EXCEPTION_EGD

Effect Type
Raise Exception

Execution Order
0

Add to Audit?
☒

Exception

Active
☒

Exception Priority
High

Reject Type
DGC Suspect Value

Assigned User

Assigned Group
Equity Management

Exception Reason

Exception Message

Exception

- **Exception priority:** defines the exception priority.
- **Reject Type:** defines the exception type.
- **Assigned User:** assigned user that will handle the exception on its exceptions list.
- **Assigned Group:** group assigned to exception that will be considered for granting rights on the exceptions management workflow.
- **Exception Reason:** currently not used.
- **Exception Message:** currently not used.

Rule Execution Tracking

The rule execution tracking feature consists of recording the results of data quality checks performed for a given object. If an anomaly is detected during the verification, the Fail status is propagated from the control clause level to the rule level.

Each rule execution (rule, control, clause) can result in one of the following statuses:

- **PASS:** no anomaly is found; it means that the assessed value does not match the target value that refers to a data issue
- **FAIL:** an anomaly is found; it means that the assessed value matches the target value that refers to a data issue
- **ERROR:** a technical error occurred during the rule execution (invalid syntax in control, server unavailable, etc)

The result of all control rule executions, along with their statuses, is stored in **Datahub Factory > Data Administration > Data Quality > DQC > Data Quality Screening Report**

PATH: HOMEPAGE > DATA QUALITY SCREENING REPORT

Data Quality Screening Report

Query ▾

Actions

1 Hidden column 2 Groups 3 Sorts

Date	Time	Object	Rule	Rule Status
27/06/2025	16:50:02	Xero Rg Registered Shs	check-secCore	Pass
16:50:02 (11)				
27/06/2025	16:50:02	Xero Rg Registered Shs	check-Ask-CS	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	Check-CS	Fail
27/06/2025	16:50:02	Xero Rg Registered Shs	Check-valueOf	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	Check-CS-Rule-Web	Fail
27/06/2025	16:50:02	Xero Rg Registered Shs	Check-High-quote	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	check-bid-quote	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	check-prices	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	check-shortName	Error
27/06/2025	16:50:02	Xero Rg Registered Shs	Check-Ask-Quote	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	Check_CashInstrument	Pass
27/06/2025	16:50:02	Xero Rg Registered Shs	check-secCore	Pass
16:41:04 (11)				
27/06/2025	16:41:04	Xero Rg Registered Shs	check-Ask-CS	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	Check-CS	Fail
27/06/2025	16:41:04	Xero Rg Registered Shs	Check-valueOf	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	Check-CS-Rule-Web	Fail
27/06/2025	16:41:04	Xero Rg Registered Shs	Check-High-quote	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	check-bid-quote	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	check-prices	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	check-shortName	Error
27/06/2025	16:41:04	Xero Rg Registered Shs	Check-Ask-Quote	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	Check_CashInstrument	Pass
27/06/2025	16:41:04	Xero Rg Registered Shs	check-secCore	Pass

383 lines

Figure 1 - Data Screening Report Screen

For each data quality performed on a given object, a new MxControlExecResult is created to carry the results of the rule execution. It contains the following data:

- **Object:** label of the object for which the quality control is performed.
- **Rule:** reference to the rule executed as part of the quality check.
- **Status:** final status of the rule execution.
- **Trigger:** refers to the action that launched the quality check. For now, only “on-demand” type of trigger is implemented (data quality check, can only be launched via business rules).
- **Control Summary:** contains the list of all controls executed as part of the quality check along with their statuses.
- **Control Details:** contains the list of all clauses verified with the following information:
 - Clause Label
 - Clause Status
 - Assessed Value
 - Target Value

PATH: HOMEPAGE > DATA QUALITY SCREENING REPORT > EXECUTION RESULT DETAILS

Execution Result Details

Q Actions

Object: Tallinna Sadam Rg Registered Shs Rule: Equity Price Check - Relative Status: Fail Trigger Type: On Demand

Controls Summary

Hide columns Group 1 Sort

Status	Control
Pass	Series Jump - Ask Quote > 10%
Pass	Series Jump - High Quote > 10%
Fail	Series Jump - Close Quote > 20%

Controls Details

3 Hidden columns Group 1 Sort

Clause	Status	Assessed Value	Target Value
Series Jump Ask Quote > Value of 0.1	Pass	0.0	0.1
Series Jump Reference close quote > Value of 0.2	Fail	0.2372881355932203	0.2
Series Jump High Quote > Value of 0.1	Pass	-0.180000000000000002	0.1

Figure 2 - Execution Results Details



System Parameters

The New DQC module requires the definition of the following system parameters (MxParameter):

- **dqc.tech.err.reject.type:** Indicates the reject type for technical errors. If the parameter is not informed, DQC_ERROR is used.
- **dqc.tech.err.reject.priority:** Indicates the reject priority for technical errors. If param is not informed, reject priority will be empty.
- **dqc.tech.err.reject.group:** Indicates the reject group for technical errors. If param is not informed, reject group will be empty.